DS3040: Introduction to Deep Learning

Dataset **PROJECT PLAN PROPOSAL** Image-Super-Resolution

TEAM 3

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PROBLEM STATEMENT

Project Name	Image Super-Resolution
Objective	The objective of this project is to solve the problem of improving low-resolution images that lack clear details due to device limits, compression, or data loss. We aim to build and evaluate deep learning models that can generate high-resolution images from low-resolution inputs while maintaining clarity, structure, and overall visual quality.

INITIAL INSIGHTS FROM DATA

Dataset Overview	Comprises 855 paired high-resolution (HR) and low-resolution (LR) RGB PNG images, predominantly featuring automotive images (cars, vehicles, riding gear) alongside other random images.
Directory Structure	Organized into train and val splits. Each split contains high_res and low_res subdirectories with identical naming, facilitating straightforward paired data loading.
Resolution Anomaly	Both HR and LR images are 256×256 pixels ($\sim$100KB each), It seems like the LR images are pre-upsampled: they were downsampled to destroy high-frequency details, then mathematically interpolated back to 256×256.

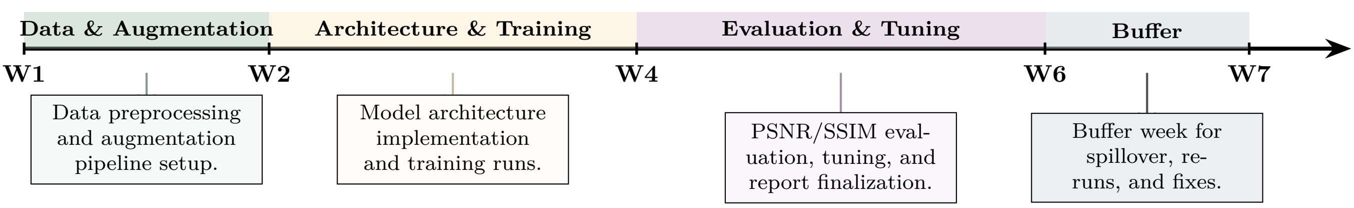
CHALLENGES WE EXPECT

Dataset Size	Severe risk of overfitting due to limited data; the dataset currently contains only 855 image pairs.
Computational	Training deep super-resolution architectures and processing large spatial tensors is highly memory-intensive and will demand significant GPU memory. We may need to find efficient strategies to manage the data load.
Evaluation	Finding the optimal balance between Distortion (minimizing the mathematical difference between pixels) and Perceptual Quality (maximizing visual sharpness), as models that achieve high pixel-wise accuracy often produce blurry results by averaging out sharp edges to avoid mathematical errors.

DIVISION OF TASKS

Aakash Kiran	Core data preprocessing pipeline, baseline architecture implementation, main training loop logic, and final quantitative synthesis.
Jeeva Jayaprakash	Dataset augmentation strategy, deep feature extraction design (e.g., residual blocks), hyper-parameter tuning, and qualitative visual benchmarking.
Muhamed Rizwan	Image degradation modeling, advanced upsampling module development (e.g., sub-pixel convolution), validation loop implementation, and experiment tracking.
Akshat Nair	Memory optimization for data loading, architectural ablation studies, advanced loss function formulation (e.g., perceptual loss), and quantitative evaluation metrics (PSNR/SSIM).

TIMELINE



THANK YOU